

Modular all drives supplier meeting preparation ▾



TODAY IM GOING TO HAVE A MEETING WITH COMAU, AN AUTOMATION COMPANY FROM ITALY TO DISCUSS ABOUT MY PROJECT MODULAR ALL DRIVES, ILL SEND YOU ATTACHED THE SCOPE THAT I CREATED FOR THIS PROJECT, THIS IS GOING TO BE THE FIRST MEETING, TO KNOW MORE ABOUT THE COMPANY AND DISCUSS SOME DETAILS ABOUT THE PROJECT AND WHAT THEY ARE ABLE TO DO ABOUT IT. ALSO THIS IS THE FIRST SUPPLIER THAT WE ARE TALKING TO, IN THE NEXT DAYS WE ARE GOING TO TALK TO OTHER SUPPLIERS LIKE, DMC from chicago, MONDRAGON, AUTEC, DEVELOP LLC, SO I NEED SOME IDEAS OF WHAT TO TALK, QUESTIONS, AND THINGS LIKE THAT, ALSO SOME ADVICES YOU HAVE TO ME.

Orchestrated meeting strategy with technical questions, commercial considerations, and multi-su... >

Great timing for this — and your spec is solid groundwork. Since this is the first of five supplier conversations, the most valuable thing you can do today is (a) gauge whether Comau is genuinely a fit, and (b) refine a standard set of questions you'll ask everyone, so you can compare suppliers fairly later. Here's how I'd approach it.

Goals for this first meeting

Keep in mind this is discovery, not negotiation. You want to learn three things: do they have real experience with *electrical test systems* (not just mechanical automation), can they support you locally in Brazil long-term, and how do they react to your spec — do they ask smart questions or just say "yes, we can do everything"? A supplier who immediately agrees to everything without probing is a red flag.

Questions about their experience and capability

- Have you built end-of-line test systems for power electronics or drives before? Can you show references or case studies? (Comau is strong in automotive body/assembly automation — test benches for VFDs with hipot, load banks, and supervisory software is a different specialty. Push on this specifically.)

- Who would develop the test software and supervisory system — Comau directly, or a subcontractor/partner? Where is that team located?
- Have you worked with hipot testing and the safety architecture around it (interlocks, discharge circuits, monitored enclosures per NR12/NR10)? Brazilian NRs are specific — ask if they've delivered NR12-compliant equipment in Brazil before.
- What's your experience integrating customer-supplied hardware? (Your spec requires WEG PLC500, WEG components and WAU-supplied electronic loads — some integrators resist using hardware they don't control. Ask how they handle warranty boundaries when the customer supplies key components.)
- Do you have engineering and field service presence in Brazil? Where? What's the typical response time for support? (Comau has a presence in Betim/MG historically — confirm what still exists and whether it covers Santa Catarina.)

Questions about how they'd approach your project

- Walk them through the concept and ask: what risks or weak points do you see in this spec? (Your own document invites the supplier to propose improvements — a good supplier will have opinions about the side-door/tray transfer mechanism, the rear docking connections at 60 A / 575 V, and the heated cabin.)
- How would they solve the automatic high-current connection (docking) reliably over thousands of cycles? Contact wear at 60 A is a classic failure point.
- How do they propose handling the product movement into the test position — gantry, shuttle, servo conveyor? (Your spec deliberately leaves this open.)
- What's their typical project methodology and timeline for a system like this — design freeze, FAT, SAT, ramp-up? Rough order of magnitude in months?
- How do they handle source code delivery? Your spec requires it — some integrators push back or deliver compiled-only. Get their position on record now.
- Virtual commissioning / digital twin capability — your spec asks for simulation or prototyping. Comau markets this heavily, so ask to see what it actually looks like in practice.

Commercial and practical questions

- Ballpark budget range for a system of this nature (they won't commit, but their reaction tells you if you're in the same universe).
- Where would the system be manufactured and assembled — Italy or Brazil? This affects cost, lead time, import taxes, FAT logistics, and spare parts.
- Warranty terms, spare parts availability locally, and training (your spec requires operational and maintenance training — ask in what language and where).
- Are they open to a phased approach (e.g., functional module first, then hipot and load) if budget requires it?

Advice for the broader process

Use the same script for everyone. Turn the questions above into a checklist and ask all five suppliers the same things. Build a simple evaluation matrix (technical capability, test-system experience, Brazil support, software competence, cost, lead time) and score each one right after the meeting while it's fresh.

Note the asymmetry in your supplier list. Comau and Mondragon are large machine builders; DMC is fundamentally a software/controls engineering firm; Autec and Develop are smaller integrators. They'll propose very different things. Decide internally whether you want a turnkey machine builder or whether you'd consider splitting mechanics and software. With DMC especially, ask who would build the mechanical side.

Listen more than you present. Spend maybe 20 minutes presenting the concept, then let them talk. The quality of their questions back to you is the single best indicator of competence. Write down what each supplier asks — questions one supplier raises become things you should clarify in the spec for the others.

Protect the spec. If you haven't already, have an NDA in place before sharing the full document — your spec reveals WEG production strategy and product details.

Be ready for the questions they'll ask you. Have answers (or honest "to be defined") for: number of product variants and production volume per shift (this drives whether cycle times are feasible), whether the three modules are one purchase order or separable, the target installation date, whether the line layout/floor space is fixed, and details of the WAU electronic loads (model, interfaces) since that's customer-supplied.

Capture spec ambiguities they expose. A few I'd anticipate: the vision system scope is "to be detailed later" — suppliers will want at least a rough definition to quote; the load test recipe (load profiles, duration, what 50 °C cabin control tolerance means) needs more definition; and clarify what "similar products" means dimensionally so fixtures aren't a moving target.

One last thing: since your document says silence equals full compliance, remind every supplier verbally of that clause and ask them to explicitly list deviations in their quotations — it's the clause that will protect you most during the proposal comparison.

Good luck in the meeting — if you'd like, after it happens I can help you turn your notes into a structured comparison matrix for the remaining suppliers.

send me a resume about comau

Identified company overview request and sourced current information >

Comau — Company Briefing

Who they are

Comau is a globally recognized leader in industrial automation, founded in 1973 — the name comes from "Consorzio Macchine Utensili." Headquartered in Turin (Grugliasco), Italy, the group has an international network of 8 innovation centers and 14 production facilities in 12 countries, with more than 4,000 employees. They were born as Fiat's in-house machinery supplier, which shaped their DNA: they are fundamentally an automotive body-shop and assembly automation house. [ZoomInfo](#)

Comau

Ownership — important context

For 50 years Comau belonged to the Fiat/FCA group and later Stellantis. In July 2024, Stellantis and One Equity Partners (OEP), a middle-market private equity firm, announced that OEP would make a majority investment in Comau, a spinoff that was part of the strategic agreement set during the FCA-PSA merger that formed Stellantis. So you're now dealing with a PE-owned company actively trying to diversify beyond automotive and win new customers — which can work in your favor commercially, but ask how the new ownership affects long-term support commitments. [Stellantis](#)

What they do

Their portfolio combines advanced robotics, vision technologies, AI, and digital platforms, including joining, assembly and machining solutions for traditional and electric vehicles, robotized manufacturing systems, industrial and collaborative robots, exoskeletons, AMRs, and end-to-end battery manufacturing and cell formation. Through Automha, a fully owned subsidiary, they also cover intralogistics and warehousing automation. They make their own industrial robot line and offer project management, consultancy, and a training Academy. [Comau + 2](#)

Their strongest reference areas are automotive body-in-white, powertrain/e-mobility assembly, and battery manufacturing — note that battery cell formation lines actually involve electrical test/charge-discharge processes, which is the closest thing in their portfolio to your test-system needs.

Presence in Brazil — relevant for you

They have a real, long-established Brazilian operation: **Comau do Brasil**, based in Betim, Minas Gerais, with entities covering engineering services (founded 1995) and machinery manufacturing. Recent local work: Comau designed and implemented a

major retooling project at the Stellantis automotive plant in Betim for production of the Fiat Pulse and Fastback SUVs. So local engineering, manufacturing, and field service exist — but they're ~1,300 km from Santa Catarina, and heavily automotive-focused. Confirm whether the Modular All Drives project would be executed by the Brazil team or from Italy. [Econodata + 2](#)

How this maps to your project — strengths and gaps to probe

Strengths: genuine global machine-building capability, strong mechanical/conveyor/robotics engineering, vision systems, digital twin/virtual commissioning experience, local Brazilian entity familiar with NR10/NR12, and global standardization experience (relevant to your "ease of global replication" requirement).

Gaps to probe: end-of-line *electrical* test systems for power electronics — hipot testers, electronic load integration, drive test software, supervisory/MES test databases — is not their core advertised business. Their world is welding, joining, and assembly. Key questions for today:

1. Show me references of test benches / EOL test systems you've delivered, especially involving high voltage or power electronics.
2. Who writes the test and supervisory software — Comau Brazil, Comau Italy, or a partner?
3. Would this project be engineered and built in Betim or in Italy? What does FAT/SAT/support look like from Betim to Blumenau/Jaraguá?
4. How comfortable are you building around customer-imposed hardware (WEG PLC500, WEG components, WAU electronic loads)?
5. Post-spinoff from Stellantis: what's the strategy for non-automotive customers in Brazil, and what guarantees of long-term support can you give?

One note of caution: as a large automotive-grade machine builder, Comau's cost structure may be heavy for a three-module test line. Their quote will be a useful "premium benchmark" against DMC, Autec, and the others — just make sure the comparison is apples-to-apples on scope, especially software and documentation deliverables.

SUPPLIERS.xlsx

XLSX

what do you think about the others suppliers? create a tier list, ill send you my list of suppliers that i sent emails and are waiting to schedule meetings

Evaluated supplier candidates and constructed tiered ranking framework >

I read your spreadsheet — nice pipeline, by the way: ~24 companies across China, Japan, Europe, USA and Brazil, with Comau today (June 11), Autec and Develop on the 15th, Mondragon on the 16th, and DMC on the 17th. Here's my honest assessment, organized as a tier list.

First, the criteria I'm ranking against, based on your spec: (1) real experience with **end-of-line electrical test systems** for power electronics (hipot, load, functional test — not just assembly automation), (2) software/test-engineering depth with source code delivery, (3) ability to support you in Santa Catarina long-term (NR10/NR12, training in Portuguese, spare parts, field service), and (4) size/cost structure compatible with a 3-module line. One disclaimer: this is a fit assessment from each company's public profile, not a judgment of quality — the meetings will confirm or destroy these rankings.

Tier S — Best theoretical fit, prioritize these meetings

Bosch (Manufacturing Solutions, Brazil) — Bosch's special-purpose machinery division builds assembly and functional test equipment for its own plants and external customers, knows electronics manufacturing intimately, is deeply fluent in NR12, and has the global standardization DNA your spec asks for. Local presence in Brazil. If they're interested in a project this size, they're arguably your strongest profile. Risk: they can be selective, slow, and premium-priced.

HAHN Automation Group (Germany) — assembly *and* functional test automation is literally their core business, including EOL testing of mechatronic and electronic products. Strong fit for the functional/load modules. Risk: support from Germany; ask about Americas footprint.

Mondragon Assembly (Spain, with Brazil unit) — automated assembly and test lines for electronics and solar (their solar lines include electrical testing stations), modular machine philosophy, and an established Brazilian operation. The combination of test experience + local support puts them high. You meet them on the 16th — good.

Tier A — Strong, worth serious meetings

BBS Automation (Germany, now part of Dürr Group) — well-regarded builder of flexible assembly and test systems, global footprint including Suzhou. Test stations are a real competence. Risk: post-acquisition integration into Dürr; no Brazil base.

ATS Corporation — (note: they're Canadian, HQ in Cambridge, Ontario — you have them under Europe). One of the largest custom automation builders in the world, with genuine test-system capability. Risk: they target very large programs; a 3-module line may be small for them, and cost will reflect their overhead.

Masmec (Italy) — Italian builder of assembly lines and EOL test benches for automotive mechatronics and e-motors. Of your European list, possibly the most test-bench-native company. Risk: no Brazil presence; mid-size company far from you.

DMC (Chicago) — different animal: an engineering services firm, not a machine builder. But they are excellent at exactly what most integrators are weakest at — test & measurement software, PLC/LabVIEW/TestStand, MES integration, traceability databases. Keep them in mind even if they don't win the whole scope: a DMC + mechanical-builder pairing, or DMC as software auditor, is a legitimate strategy. Ask in your meeting on the 17th who they'd partner with for the mechanics.

Autec (Joinville) — I can't fully judge their technical depth in hipot/load testing, but being in your own city is an enormous practical advantage for commissioning, support, spare parts, training, and lifetime cost. If their test-system competence checks out in Monday's meeting, they jump to Tier S. Probe hard on: largest test system delivered, software team size, high-voltage experience.

PIA Automation (Suzhou) — PIA is an international group (Austria/Germany/China) doing assembly and test automation including e-mobility EOL test. Technically credible. Risk: support chain from China/Europe to Brazil.

Tier B — Plausible, but with significant gaps

Comau / FFT Produktionssysteme — both are automotive body-shop and assembly giants. World-class mechanically, but EOL electrical test for power electronics is not their natural habitat, and their cost structures are built for automotive programs. Useful benchmarks; I'd be surprised if either is the best-value winner.

JR Automation (Hitachi) / Eclipse Automation (Accenture) — big, competent North American integrators with broad portfolios including some test. Similar concern: generalists, premium overhead, no Brazil support base.

Harmontronics (China) — large listed Chinese intelligent-manufacturing company, strong in battery/3C assembly automation. Capability likely exists, but remote support, documentation language, and NR12 compliance are real hurdles.

Aumaq, SPI, MCK (Brazil) — Brazilian machine builders/integrators. Local support is a plus and pricing may be attractive, but verify whether they've ever built anything at this complexity level (automated docking at 60 A / 575 V, hipot safety architecture, supervisory + MES integration, heated cabin). For some of them this could be a career-defining stretch project — which can mean great dedication or great risk.

Acieta (USA) — primarily a FANUC robotics integrator focused on material handling and machine tending. Your project has little robotics and a lot of electrical test — probable mismatch.

Develop LLC (Wisconsin) — a single-source automation company based in Verona, Wisconsin, specializing in custom automation with embedded systems and robotics, recently a FANUC Authorized System Integrator, with in-house fabrication and FAT at their facility. Looks like a solid regional Midwest machine builder — but a small US integrator supporting a high-voltage test line in Santa Catarina (SAT, warranty, NR12, training in Portuguese) is a stretch. In Monday's meeting, the decisive question is: have you ever installed and supported anything outside the US? financialcontent Develop

Tier C — Likely wrong type of company (verify before investing meeting time)

iAutomation — to my knowledge this is a US *distributor* of automation components (motion control, sensors), not a systems integrator. Confirm before scheduling.

The Morse Group — electrical contracting / industrial services group; not an EOL test machine builder.

Tamadic (Japan) — primarily an engineering design-services firm (heavily automotive design outsourcing), not a turnkey test-machine builder.

CASMT, Ronstein, Tokyo Densai, Tokyo area / minor Chinese integrators — I genuinely don't have reliable information on these. Before giving them meeting slots, ask each for 2–3 references of delivered EOL test systems with photos/videos. That single email filters fast.

Three strategic observations

Your funnel is heavy on assembly-automation companies and light on test-system specialists. Companies like Comau, FFT, JR, Eclipse, Acieta build things that *move and join parts*; your project is mostly about things that *energize and measure products*. If you can still add names, look for dedicated EOL test bench builders for power electronics/appliances — that profile (common in Germany, Italy, and the auto-electronics supply chain) matches your scope more naturally than body-shop giants.

Define your support requirement early, and let it kill candidates fast. Decide internally what's the maximum acceptable response time for a line-down situation. If the answer is "48h on site," half this list eliminates itself, and you save weeks of meetings.

Use the strong locals as your anchor. A realistic winning outcome for you might be: a Tier S international (Bosch, HAHN, Mondragon) for credibility and test know-how, benchmarked against Autec or another capable Brazilian for cost and proximity —

with the spec's "silence = full compliance" clause enforced ruthlessly when comparing quotes.

If you want, after your meetings this week I can help you build a weighted scoring matrix (xlsx) with all these companies and your real impressions, so the decision becomes traceable for your management.

QUANTAS FABRICAS A WEG TEM NO MUNDO?

Reconciliou fontes divergentes sobre fábricas globais da WEG >

Boa pergunta — e curioso você perguntar, já que é "de casa". Os números variam um pouco conforme a fonte e o critério (parque fabril vs. unidade fabril), mas o panorama atual é este:

- No Brasil, a WEG possui 14 parques fabris, com sede e principais unidades em Jaraguá do Sul (SC), e demais fábricas no Amazonas (Manaus), Espírito Santo (Linhares), Minas Gerais (Betim), Rio Grande do Sul (Bento Gonçalves e Gravataí), Santa Catarina (Blumenau, Guaramirim e Itajaí) e São Paulo (São Paulo, São Bernardo do Campo, Sertãozinho, Mauá e Monte Alto); [Wikipedia](#)
- No exterior, são 48 unidades fabris; [Wikipedia](#)
- Somando, isso dá cerca de **60+ unidades fabris no mundo**, distribuídas em 17 países. [OCP News](#)

Para contexto geral: a empresa tem operações em cerca de 140 países e aproximadamente 40.000 funcionários, com presença comercial e de distribuição em dezenas de outros mercados além dos países onde fabrica. [Wikipedia](#)

Vale dizer que esse número muda com frequência — a WEG vem crescendo por aquisições (Regal Rexnord industrial motors, Volt em Türkiye, etc.), então plantas novas entram no mapa quase todo ano. Se você precisar do número oficial exato para algum material de apresentação, o melhor é conferir o Relatório Anual Integrado ou a página de RI da WEG, que trazem o dado consolidado e auditado.

Aliás, isso conecta direto com o seu projeto: o requisito de "replicação global" do Modular All Drives faz ainda mais sentido visto desse ângulo — vale mencionar aos fornecedores que o conceito pode ser replicado em outras plantas de drives do grupo no exterior, porque isso muda o apetite comercial deles (projeto de 3 módulos hoje, potencial de linha global amanhã).

Hello Aluizio

Thanks for contacting PIA. I am the general manager for PIA Suzhou China. Pls send us the presentation, we will check whether we are capable of that, if so, lets schedule a meeting to discuss.

Thanks!

answer this email for me

Orchestrated professional email response with spec attachment and meeting proposal >

Subject: RE: Modular All Drives Test System – Technical Specification & Meeting Proposal

Dear [Name],

Thank you for your prompt reply and for your interest in our project.

Please find attached the technical specification of the Modular All Drives Test System. In summary, the scope covers the design, manufacturing and installation of three automated test modules (hipot test, functional test and load test) for end-of-line validation of WEG variable frequency drives, including integrated conveyors, automatic electrical connections, test and supervisory software, and full compliance with the applicable safety standards (NR10/NR12, IEC).

A few key points for your initial capability assessment:

- End-of-line electrical testing of power electronics: hipot (dielectric), functional and load testing up to 60 A at 220/440/575 V;
- Automated product handling and docking connections, with one test position per module;
- Test software and supervisory software development, with source code supply and MES/traceability integration;
- Installation, commissioning (FAT/SAT), training and after-sales support at our facility in Jaraguá do Sul, Santa Catarina, Brazil.



Send via Gmail



Here's your reply, ready to send. A few notes on the choices I made:

- **It summarizes the scope in 4 bullets** before they even open the PDF — this lets the GM do a fast go/no-go internally, which is exactly what he asked for ("we will check whether we are capable");
- **It front-loads the two filters that matter most for a China-based supplier:** the support/commissioning location (Jaraguá do Sul, Brazil) and the source code requirement. If either is a dealbreaker for them, better to find out now than after three meetings;
- **It proposes a concrete time window** accounting for the ~11-hour difference (morning in Brazil = evening in Suzhou), which speeds up scheduling;

- **It includes a confidentiality line** — since you likely don't have an NDA with them yet, this at least puts the expectation in writing. If WEG procurement has a standard NDA template, even better: you could attach it and condition the spec on its signature instead.

Just replace **[Name]** with the GM's name from his signature, attach the PDF, and adjust the signature block if your official title differs. If you'd rather hold the spec until an NDA is signed, tell me and I'll redraft it in that more cautious version.

responda de uma forma mais simples e direta e diga que o escopo estará em anexo

Subject: RE: Modular All Drives Test System – Scope Attached

Dear [Name],

Thank you for your reply.

Please find attached the scope of the Modular All Drives Test System project. It covers three automated test modules (hipot, functional and load test) for end-of-line testing of WEG variable frequency drives, to be installed at our facility in Jaraguá do Sul, Brazil.

Once you have reviewed the document, please let me know if PIA is capable of this scope so we can schedule a meeting. Considering the time difference, mornings in Brazil (evenings in China) work best for us.

This document is confidential and for evaluation purposes only.

Best regards,

Aluizio Augusto Kleine Kirschner
Industrial Engineering – Drives Production Department
WEG Drives & Controls – Automation Ltd.



Send via Gmail

Pronto — versão bem mais enxuta: agradece, diz que o escopo está em anexo com um resumo de uma frase, pede que confirmem a capacidade para agendar a reunião, e mantém só uma linha de confidencialidade. É só trocar o **[Name]** pelo nome do gerente e anexar o PDF.

Hi Aluizio,

Thanks for reaching out to JR Automation.

I received your message regarding automated test systems for a frequency inverter line. I did not see an attachment illustrating project specifications. Where would the location of this project be?

answer this email saying that im going to attach the document and the location will be brazil

Subject: RE: Automated Test Systems – Specification Attached / Project Location: Brazil

Hi [Name],

Thank you for your reply.

Apologies for the missing file — the project specification is now attached. It covers three automated test modules (hipot, functional and load test) for end-of-line testing of WEG variable frequency drives.

The project will be installed at our facility in Jaraguá do Sul, Santa Catarina, Brazil.

Once you have reviewed the document, please let me know if this scope fits JR Automation's capabilities so we can schedule a meeting to discuss it further.

This document is confidential and for evaluation purposes only.

Best regards,

Aluizio Augusto Kleine Kirschner
Industrial Engineering – Drives Production Department
WEG Drives & Controls – Automation Ltd.



Send via Gmail



Pronto — no mesmo estilo direto da anterior: confirma o anexo, responde a pergunta da localização (Jaraguá do Sul, SC, Brasil) e já abre caminho para agendar a reunião. É só trocar o **[Name]** e anexar o PDF.

Uma dica: a pergunta sobre localização não é casual — para integradores americanos como a JR, instalação e suporte no Brasil é o principal fator de decisão sobre participar ou não. Se eles seguirem adiante, espere perguntas sobre quem faria comissionamento e suporte local; vale você já ter essa resposta preparada (por exemplo, se a WEG aceitaria suporte remoto + visitas programadas, ou exige presença local).

